Qassim University
جامعة القصيم
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مُبصر

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1- INTRODUCTION

PROBLEM

Current assistive technologies suffer from limited services and high latency, failing to provide a comprehensive understanding of dynamic environments. Visually impaired users face significant risks due to the lack of integrated tools that can simultaneously detect obstacles and recognize text in real-time.

SOLUTION

Mubser bridges these gaps by unifying Object Detection, Obstacle Detection, and OCR into a single, AI-powered real-time pipeline. The system captures environmental cues and delivers instant audio feedback, ensuring safe navigation and text recognition with minimal cognitive load for the user.

2- METHODOLOGY

Stage 1 :

Image Acquisition

The process begins with capturing real-time visual data via the device's camera. This serves as the input for the system, providing a continuous representation of the user's surroundings.

1

Stage 2 :

Object Detection and Obstacle Identification

In this stage, the system employs Computer Vision algorithms to perform Object Detection.
It identifies physical Obstacles and various entities in the environment to facilitate safe navigation through spatial analysis.

2

Stage 3 :

Optical Character Recognition (OCR)

The system analyzes the captured frames to detect and extract textual information using OCR technology. This enables the recognition of signs, labels, and documents, converting visual text into processed digital data.

3

Stage 4 :

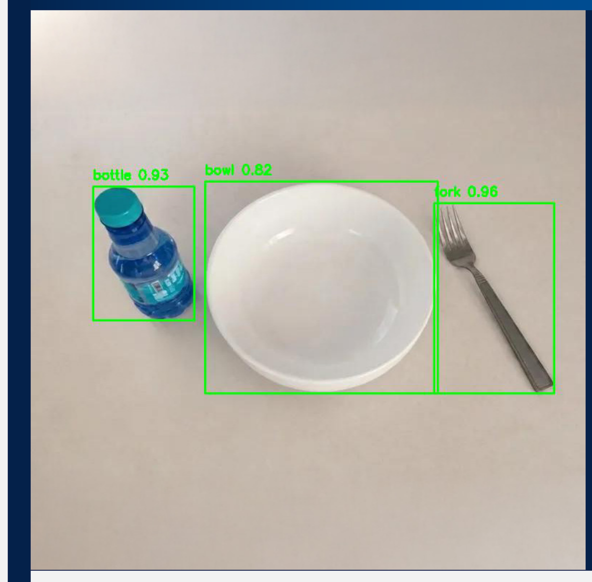
Voice Output Generation

In the final stage, the analyzed information is synthesized into natural language. The system generates Voice Output to provide the user with immediate and descriptive audio feedback regarding their environment.

4

3- EXPERIMENTS

OBJECT DETECTION



Mubser:
3 objects have been detected: bottle, bowl, fork.

يوجد أمامك علبة، وعاء، وشوكة.

Obstacles Detection

Mubser:
1 obstacle has been detected: bad road.

انتبه! يوجد أمامك طريق سيئ.

OCR

Mubser:
1 object has been detected: book.

يوجد أمامك كتاب النص المكتشف:"دان براون - شيفرة دافنشي"

Detected text:
"Dan Brown - The Da Vinci Code"

4- CONCLUSION

Mubser successfully integrates Object Detection, Obstacle Detection, and OCR into a single efficient pipeline. The system provides reliable real-time environmental awareness, significantly enhancing independence and safety for visually impaired users. Our results confirm that AI-driven assistive technology is a vital step toward a more accessible and inclusive future.

Capture visual input (Input Device)
↓
Identify object & detect text (Object Detection)
↓
Text detected?
Yes → Extract text from image (OCR)
No → Send object information
↓
Text Preprocessing & Formatting
↓
Generate spoken output (Audio Feedback)

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